

MH1200 Linear Algebra I.

Problem set #9.

SOLUTIONS

Problem 1:

Let V be the subset of $\mathbb{R}^4$ given by

$$V = \{ (w, x, y, z) \mid w, x, y, z \in \mathbb{R}, \quad w + x = y + z = -x + 2y \}.$$

1. Show that V is a subspace of $\mathbb{R}^4$.
2. Express V as the span of a collection of vectors in $\mathbb{R}^4$.

Solution to Part 1:

The condition $w + x = y + z = -x + 2y$ is the same as the pair of equations $w + x = y + z$ and $y + z = -x + 2y$. These can be rewritten as a homogeneous linear system:

$$\begin{aligned} w + x - y - z &= 0 \\ x - y + z &= 0 \end{aligned}$$

Then the solution set is a subspace according to a standard theorem we proved in the lectures.

Solution to Part 2:

The general solution of the homogeneous linear system above is easily found by back-substitution, taking $y = s$ and $z = t$ as free parameters. (There is no need to perform Gaussian elimination since the augmented matrix is already in row-echelon form in this case.) The result is:

$$(w, x, y, z) = (2t, s - t, s, t) = s(0, 1, 1, 0) + t(2, -1, 0, 1).$$

Since s and t can have any values, this shows that

$$V = \text{span}\{ (0, 1, 1, 0), (2, -1, 0, 1) \}.$$

□

Problem 2:

Consider the homogeneous linear system

$$\begin{aligned}10x_2 - 4x_3 + x_4 &= 0 \\x_1 + 4x_2 - x_3 + x_4 &= 0 \\3x_1 + 2x_2 + x_3 + 2x_4 &= 0 \\-2x_1 - 8x_2 + 2x_3 - 2x_4 &= 0 \\x_1 - 6x_2 + 3x_3 &= 0\end{aligned}$$

Express the solution set as the span of some vectors in $\mathbb{R}^4$.

Solution

The general solution is found by Gaussian elimination and back-substitution. The result of the calculations is:

$$\begin{aligned}(x_1, x_2, x_3, x_4) &= \left(-\frac{3}{5}s - \frac{3}{5}t, \frac{2}{5}s - \frac{1}{10}t, s, t\right) \quad s, t \in \mathbb{R} \\&= s\left(-\frac{3}{5}, \frac{2}{5}, 1, 0\right) + t\left(-\frac{3}{5}, -\frac{1}{10}, 0, 1\right)\end{aligned}$$

Hence the solution set can be expressed as $\text{span}\left\{\left(-\frac{3}{5}, \frac{2}{5}, 1, 0\right), \left(-\frac{3}{5}, -\frac{1}{10}, 0, 1\right)\right\}$.

□

Problem 3:

Consider the following collection of vectors in $\mathbb{R}^4$:

$$S = \{(0, 1, 0, 0), (-a, 0, 1, -1), (0, a, 0, 0), (1, 0, 0, 1)\}$$

where a is a real number.

- (i) Show that no matter what we set a to, $\text{span}(S) \neq \mathbb{R}^4$.
- (ii) Find an example of a vector in $\mathbb{R}^4$ that is not contained in $\text{span}(S)$.

Solution

We can solve both of these problems at once. Let an arbitrary vector in $\mathbb{R}^4$ be denoted (p, q, r, s) . To see if this vector lies in the given span we have to solve the usual equation:

$$(p, q, r, s) = c_1(0, 1, 0, 0) + c_2(-a, 0, 1, -1) + c_3(0, a, 0, 0) + c_4(1, 0, 0, 1).$$

The augmented matrix of this linear system in the variables c_1, c_2, c_3, c_4 is

$$\left[\begin{array}{cccc|c} 0 & -a & 0 & 1 & p \\ 1 & 0 & a & 0 & q \\ 0 & 1 & 0 & 0 & r \\ 0 & -1 & 0 & 1 & s \end{array} \right]$$

Now we solve with Gaussian elimination:

$$\left[\begin{array}{cccc|c} 1 & 0 & a & 0 & q \\ 0 & 1 & 0 & 0 & r \\ 0 & -a & 0 & 1 & p \\ 0 & -1 & 0 & 1 & s \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & a & 0 & q \\ 0 & 1 & 0 & 0 & r \\ 0 & 0 & 0 & 1 & p+ar \\ 0 & 0 & 0 & 1 & s+r \end{array} \right] \rightarrow \left[\begin{array}{cccc|c} 1 & 0 & a & 0 & q \\ 0 & 1 & 0 & 0 & r \\ 0 & 0 & 0 & 1 & p+ar \\ 0 & 0 & 0 & 0 & s-p+(a-1)r \end{array} \right]$$

Looking at this, we see that if we set $p = q = r = 0$ and $s = 1$ then the system specialises to

$$\left[\begin{array}{cccc|c} 1 & 0 & a & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

The last row of this system is of the inconsistent form, so there will be no solutions. So the corresponding vector

$$(0, 0, 0, 1)$$

cannot be expressed as a linear combination of the given vectors.

□

Problem 4:

Consider the following vectors in $\mathbb{R}^4$:

$$\mathbf{u}_1 = (2, 0, 2, -4), \mathbf{u}_2 = (1, 0, 2, 5), \mathbf{u}_3 = (0, 3, 6, 9), \mathbf{u}_4 = (1, 1, 2, -1), \\ \mathbf{v}_1 = (-1, 2, 1, 0), \mathbf{v}_2 = (3, 1, 4, 0), \mathbf{v}_3 = (0, 1, 1, 3), \mathbf{v}_4 = (-4, 3, -1, 6).$$

Determine if the following are true.

- (i) $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.
- (ii) $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \subseteq \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.
- (iii) $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} = \mathbb{R}^4$.
- (iv) $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} = \mathbb{R}^4$.

Solution to Part (i).

In lectures we learnt how to compare spans. The way to test

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

is to test whether each “building block” of the LHS $\mathbf{u}_i$ can be “built” using the “building blocks” of the RHS. Namely:

$$\mathbf{u}_i \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}.$$

We learnt a matrix algorithm for checking whether a vector lies in a span like this. The starting point is to make a matrix with the first set of columns coming from the $\mathbf{v}_i$'s and the second set of columns coming from the vectors we are testing, the $\mathbf{u}_i$'s. We write down:

$$\left[\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 2 & 1 & 1 & 3 & 0 & 0 & 3 & 1 \\ 1 & 4 & 1 & -1 & 2 & 2 & 6 & 2 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array} \right]$$

Now we do Gaussian elimination:

$$\rightarrow \left[\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 7 & 1 & -5 & 4 & 3 & 6 & 3 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \end{array} \right]$$

Continuing:

$$\rightarrow \left[\begin{array}{cccc|cccc} -1 & 3 & 0 & -4 & 2 & 1 & 0 & 1 \\ 0 & 7 & 1 & -5 & 4 & 2 & 3 & 3 \\ 0 & 0 & 3 & 6 & -4 & 5 & 9 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 & 0 \end{array} \right]$$

This is in row echelon form so we can now interpret our answer. To decide if one of the $\mathbf{u}_i$'s can be written as a linear combination of the $\mathbf{v}_i$'s we forget all columns on the right except the column corresponding to the particular $\mathbf{u}_i$ we are thinking about. If the resulting system is consistent then that $\mathbf{u}_i$ is a linear combination of the $\mathbf{v}_i$'s. If it is inconsistent then it cannot.

Looking at this matrix we deduce:

- $\mathbf{u}_1, \mathbf{u}_4 \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$
- $\mathbf{u}_2, \mathbf{u}_3 \notin \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

Thus the answer to this part is FALSE.

Solution to Part (ii).

In this part we do exactly the same thing but swapping the roles of the $\mathbf{u}_i$ and the $\mathbf{v}_i$. The computation proceeds:

$$\left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 6 & 2 & 1 & 4 & 1 & -1 \\ -4 & 5 & 9 & -1 & 0 & 0 & 3 & 6 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 7 & 9 & 1 & -2 & 6 & 3 & -2 \end{array} \right]$$

Continuing:

$$\rightarrow \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & -33 & -6 & -16 & -1 & -4 & -23 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} 2 & 1 & 0 & 1 & -1 & 3 & 0 & -4 \\ 0 & 1 & 6 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 3 & 1 & 2 & 1 & 1 & 3 \\ 0 & 0 & 0 & 5 & 6 & 10 & 7 & 10 \end{array} \right]$$

This matrix is in row echelon form. The interpretation of this matrix is that because the part of the matrix to the left of the partition has no all zero rows, the four vectors $\mathbf{u}_1, \dots, \mathbf{u}_4$ actually span all of $\mathbb{R}^4$. In particular the four vectors $\mathbf{v}_1, \dots, \mathbf{v}_4$ certainly can be built from the $\mathbf{u}_i$, so we conclude that in this case the proposition is TRUE:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \subset \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$$

Solution to Part (iii).

In Part (ii) we noticed that when we make a matrix with the vectors $\mathbf{u}_i$ as columns, then the row echelon form has no row of zeros. According to the algorithm we developed in lectures, this means that indeed

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\} = \mathbb{R}^4.$$

Solution to Part (iv).

In Part (i) we calculated that the vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ do not lie in the span of the $\mathbf{v}_i$. So we know already that

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\} \neq \mathbb{R}^4.$$

Thus the given statement in this part is FALSE.

□

Problem 5:

Determine which of the following statements are true. Justify your answer.

- (i) If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are vectors in $\mathbb{R}^3$ such that $\mathbf{u} \neq \mathbf{v}$, $\mathbf{u} \neq \mathbf{w}$ and $\mathbf{v} \neq \mathbf{w}$, then $\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \mathbb{R}^3$.
- (ii) If S_1 and S_2 are two subsets of a real vector space, then $\text{span}(S_1 \cap S_2) = \text{span}(S_1) \cap \text{span}(S_2)$.
- (iii) If S_1 and S_2 are two subsets of a real vector space, then $\text{span}(S_1 \cup S_2) = \text{span}(S_1) \cup \text{span}(S_2)$.

Solution to (i)

This is FALSE. For example:

$$\mathbf{u} = (1, 0, 0), \mathbf{v} = (0, 1, 0), \mathbf{w} = (1, 1, 0).$$

These three vectors are all different from each other but all have a zero in the third position, so there is no way they can span all of $\mathbb{R}^3$ because any linear combination of them will have a zero in the third position.

Solution to (ii)

Again this is FALSE. An example in $\mathbb{R}^2$ might be the following. Consider

$$S_1 = \{(1, 0), (0, 1)\}, S_2 = \{(1, 1), (1, -1)\}.$$

Note that

$$\text{span}(S_1 \cap S_2) = \text{span}(\emptyset) = \{0\}.$$

But

$$\text{span}(S_1) \cap \text{span}(S_2) = \mathbb{R}^2 \cap \mathbb{R}^2 = \mathbb{R}^2.$$

(These sets both span $\mathbb{R}^2$. We proved this in lectures and it is easy to check.)

Solution to (iii)

This is obviously FALSE. The RHS isn't even a subspace except in special cases. For example if $S_1 = \{(1, 0)\}$ and $S_2 = \{(0, 1)\}$ then the LHS is $\text{span}\{(1, 0), (0, 1)\} = \mathbb{R}^2$ while the RHS is just the coordinate axes.

□

Problem 6:

Let V be the subset of $\mathbb{R}^4$ given by

$$V = \{(a, b, c, d) \mid a, b, c, d \in \mathbb{R} \text{ and } a + b = c + d\}.$$

- (i) Show that V is a subspace of $\mathbb{R}^4$.

(ii) Let $S = \{(1, 0, 2, -1), (0, -3, 1, -4), (3, 6, 4, 5)\}$.

Is $\text{span}(S) \subseteq V$? Is $V \subseteq \text{span}(S)$? Is $\text{span}(S) = V$?

Solution to (i)

This set is obviously the set of solutions to a homogeneous linear system $a + b - c - d = 0$. Thus by a standard theorem it is definitely a subspace.

Solution to (ii)

The vectors $(a, b, c, d) = (1, 0, 2, -1), (0, -3, 1, -4), (3, 6, 4, 5)$ all satisfy the condition $a + b = c + d$ and are therefore vectors in V . Since V is a subspace, it follows that all linear combinations of these vectors are contained in V , and this means that $\text{span}(S) \subseteq V$.

To see if $V \subseteq \text{span}(S)$: One way to do this would be to find a set of spanning vectors for V and then to check whether they can all be expressed as linear combinations of the vectors in S .

Here for variety we'll take a slightly different approach to this problem. Note first that the condition $a + b = c + d$ is the same as $d = a + b - c$, so all vectors in V have the form $(a, b, c, a + b - c)$ for arbitrary $a, b, c \in \mathbb{R}$. The question now is whether every vector of this form is contained in $\text{span}(S)$, i.e. can be written as a linear combination of the vectors in S . But this is equivalent to having a solution of the linear system

$$\begin{pmatrix} 1 & 0 & 3 \\ 0 & -3 & 6 \\ 2 & 1 & 4 \\ -1 & -4 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \\ a + b - c \end{pmatrix}$$

(The columns in the matrix are the vectors in S .) To find out if a solution exists, consider the augmented matrix and reduce it to row-echelon form. The result is

$$\left(\begin{array}{ccc|c} 1 & 0 & 3 & a \\ 0 & -3 & 6 & b \\ 2 & 1 & 4 & c \\ -1 & -4 & 5 & a + b - c \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 3 & a \\ 0 & 1 & -2 & -2a + c \\ 0 & 0 & 0 & -6a + b + 3c \\ 0 & 0 & 0 & -6a + b + 3c \end{array} \right).$$

Thus in this case the system is consistent only when $-6a + b + 3c = 0$. But there exist (many) values of a, b, c for which this is nonzero. For example $a = 1, b = c = 0$ gives $-6a + b + 3c = -6 \neq 0$. So the system is not consistent for all values of a, b, c and this means that not all vectors in V can be written as linear combinations of the vectors in S . Hence it is not true that $V \subseteq \text{span}(S)$. Therefore it is also not true that $V = \text{span}(S)$.

□

Problem 7:

(Version 2.)

Consider vectors

$$\vec{v}_1 = (1, 0, -1, 1), \vec{v}_2 = (1, 1, 1, 1), \vec{v}_3 = (3, 2, a, -2),$$

and vectors

$$\vec{w}_1 = (0, 1, 0, 5), \vec{w}_2 = (4, 3, 0, -1).$$

Note that these vectors depend on a parameter a .

For what values of the parameter a is it true that

$$\text{span}\{\vec{w}_1, \vec{w}_2\} \subset \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}.$$

Solution

According to the theory for comparing spans that we met in lectures,

$$\text{span}\{\vec{w}_1, \vec{w}_2\} \subset \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$$

will be true precisely when both of the vectors $\vec{w}_1$ and $\vec{w}_2$ are linear combinations of the vectors $\vec{v}_1$, $\vec{v}_2$ and $\vec{v}_3$.

We derived how to check that in lectures. We build an augmented matrix whose columns to the left of the partition are the vectors $\vec{v}_1$, $\vec{v}_2$ and $\vec{v}_3$, and whose two columns to the right of the partition are the vectors $\vec{w}_1$ and $\vec{w}_2$. That is:

$$\left[\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ -1 & 1 & a & 0 & 0 \\ 1 & 1 & -2 & 5 & -1 \end{array} \right].$$

Recall that this augmented matrix expresses the two systems of linear equations

$$c_1(1, 0, -1, 1) + c_2(1, 1, 1, 1) + c_3(3, 2, a, -2) = (0, 1, 0, 5)$$

and

$$c_1(1, 0, -1, 1) + c_2(1, 1, 1, 1) + c_3(3, 2, a, -2) = (4, 3, 0, -1).$$

To be precise: if we forget the last column we get the first of these systems. If, on the other hand, we forget the second-last column we get second of these systems. Because the part to the left of the partition is the same we can solve both systems at once, which is why we write it like this.

Now we need to see for which values of a we have a solution (i.e. a consistent system). So we do Gaussian elimination:

$$\rightarrow \left[\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 2 & a+3 & 0 & 4 \\ 0 & 0 & -5 & 5 & -5 \end{array} \right] \rightarrow \left[\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & a-1 & -2 & -2 \\ 0 & 0 & -1 & 1 & -1 \end{array} \right]$$

Continuing:

$$\left[\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & a-1 & -2 & -2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|cc} 1 & 1 & 3 & 0 & 4 \\ 0 & 1 & 2 & 1 & 3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & a-3 & -1-a \end{array} \right].$$

The first system of equations is obtained by forgetting the last column to get:

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & a-3 \end{array} \right].$$

This is consistent precisely when $a = 3$.

The second system of equations is obtained by forgetting the second-last column to get

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & -1-a \end{array} \right].$$

This system is consistent only when $a = -1$.

In conclusion: there is no value of a for which both of these systems are consistent, so there is no value of a for which

$$\text{span}\{\vec{w}_1, \vec{w}_2\} \subset \text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}.$$

□

Problem 8:

Consider some subspace $V \subset \mathbb{R}^n$ and assume that V is spanned by vectors

$$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}.$$

(i) Show that the following is also a spanning set

$$\{\mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_2 - \mathbf{v}_3, \dots, \mathbf{v}_{m-1} - \mathbf{v}_m, \mathbf{v}_m\}.$$

(ii) Show that the following is also a spanning set

$$\{\mathbf{v}_1, \mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3, \dots, \mathbf{v}_1 + \dots + \mathbf{v}_m\}.$$

Solution to (i).

To show that the given vectors are also a spanning set for $V = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ we need to show that every one of the vectors $\mathbf{v}_i$ can be expressed in terms of these vectors. Note that for the vectors $\mathbf{v}_i$ with $1 \leq i < m$ one writes

$$\mathbf{v}_i = (\mathbf{v}_i - \mathbf{v}_{i+1}) + (\mathbf{v}_{i+1} - \mathbf{v}_{i+2}) + \dots + (\mathbf{v}_{m-1} - \mathbf{v}_m) + \mathbf{v}_m$$

and the remaining vector is just $\mathbf{v}_m$ which is already on the list. (A formal proof of these identities could be written down via induction.)

Solution to (ii).

In this case the expression you need are, for all $1 < i \leq m$:

$$\mathbf{v}_i = (\mathbf{v}_1 + \dots + \mathbf{v}_i) - (\mathbf{v}_1 + \dots + \mathbf{v}_{i-1}).$$

The remaining vector $\mathbf{v}_1$ is already on the list.

□

Problem 9:

(from a 2020 midterm.)

Consider the following two subsets of $\mathbb{R}^4$ depending on a parameter a :

$$S_1 = \{(1, 2, 2, 3), (0, 3, 1, -1)\}.$$

$$S_2 = \{(1, -1, 1, 0), (-1, 1, -1, a)\}.$$

- (a) Are there any values of a where either of the vectors in S_2 lie in the span of S_1 ?
- (b) In this part set $a = 2$. Find a finite set of vectors $S_3 \subset \mathbb{R}^4$ such that

$$\text{span}(S_3) = \text{span}(S_1) \cap \text{span}(S_2).$$

Please show all the steps in your working.

Solution to (a).

To decide if either of the vectors in S_2 lie in the span of S_1 we can follow the matrix algorithm we learned at the end of the “span” topic. We begin by writing down the following matrix. The first two columns are the two vectors of S_1 , and the two columns to the right of the partition are the two vectors of S_2 .

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 2 & 3 & -1 & 1 \\ 2 & 1 & 1 & -1 \\ 3 & -1 & 0 & a \end{array} \right]$$

Then we do Gaussian elimination:

$$\rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 3 & -3 & 3 \\ 0 & 1 & -1 & 1 \\ 0 & -1 & -3 & a+3 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -4 & a+4 \end{array} \right].$$

This matrix is in row echelon form.

Now, to test whether the first vector of S_2 lies in the span of S_1 we cover over the final column of this matrix (because that column corresponds to the second vector of S_2). The result is an inconsistent system, so the vector $(1, -1, 1, 0)$ cannot be expressed as a linear combination of the two vectors in S_1 .

Similarly, to test the second vector of S_2 , we cover over the third column of this matrix. There are two possibilities.

- If $a = -4$ then the system is consistent, and the vector $(-1, 1, -1, a)$ does indeed lie in the span of S_1 .
- For every other choice of a the system is inconsistent, and neither vector will lie in the span of S_1 .

Solution to (b).

The next part is to determine a finite list of vectors S_3 such that

$$\text{span}(S_3) = \text{span}(S_1) \cap \text{span}(S_2).$$

Let us first determine a parametrization of the intersection

$$\text{span}(S_1) \cap \text{span}(S_2).$$

We can do this by first writing down an arbitrary vector of $\text{span}(S_2)$. It will be of the form

$$r(1, -1, 1, 0) + s(-1, 1, -1, 2)$$

for some real numbers r and s .

Now we calculate: for what values of r and s does this vector lie in the span of S_1 ? That will give us a parametrization of the intersection.

To answer this question we can just apply the same matrix algorithm directly to the vector

$$r(1, -1, 1, 0) + s(-1, 1, -1, 2) = (r - s, -r + s, r - s, 2s).$$

The matrix algorithm tells us to write down the following matrix depending on the two parameters r and s . The first two columns are the vectors of S_1 and the third column is this arbitrary vector from the span of S_2 .

$$\left[\begin{array}{cc|c} 1 & 0 & r - s \\ 2 & 3 & -r + s \\ 2 & 1 & r - s \\ 3 & -1 & 2s \end{array} \right].$$

Now we do Gaussian elimination:

$$\rightarrow \left[\begin{array}{cc|c} 1 & 0 & r - s \\ 0 & 3 & -3r + 3s \\ 0 & 1 & -r + s \\ 0 & -1 & 2s - 3(r - s) \end{array} \right] = \left[\begin{array}{cc|c} 1 & 0 & r - s \\ 0 & 1 & -r + s \\ 0 & 1 & -r + s \\ 0 & -1 & -3r + 5s \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & r - s \\ 0 & 1 & -r + s \\ 0 & 0 & 0 \\ 0 & 0 & -4r + 6s \end{array} \right]$$

This matrix is consistent exactly when

$$-4r + 6s = 0.$$

We can express all possible (r, s) solving this equation by setting s to a free parameter a and then $r = \frac{3}{2}a$.

We have determined the intersection $\text{span}(S_1) \cap \text{span}(S_2)$ to be

$$\left\{ \frac{3}{2}a(1, -1, 1, 0) + a(-1, 1, -1, 2); a \in \mathbb{R} \right\}.$$

This equals:

$$\{c(1, -1, 1, 4); c \in \mathbb{R}\}.$$

(Handwritten: $0.5a, -0.5a, 0.5a, 2a$)

So

$$S_3 = \{(1, -1, 1, 4)\}.$$

(Handwritten: $0.5a(1, -1, 1, 4)$)

Comments of the grader from 2020.

The most common mistake in the first part was putting the respective vectors on the wrong side of the partition. That is, mistakenly starting with:

$$\left[\begin{array}{cc|cc} 1 & -1 & 1 & 0 \\ -1 & 1 & 2 & 3 \\ 1 & -1 & 2 & 1 \\ 0 & a & 3 & -1 \end{array} \right]$$

In the second part:

The most frequent mistake was to assume that

$$\text{span}(S_1) \cap \text{span}(S_2) = \text{span}(S_1 \cap S_2).$$

This example in fact illustrates that this fact is not true.

A nice alternative approach used by some students was to express the intersection as solutions of the equation

$$x(1, -1, 1, 0) + y(-1, 1, -1, 2) = z(1, 2, 2, 3) + w(0, 3, 1, -1).$$

A number of students correctly wrote down and solved this equation but after that failed to interpret the resulting set of solutions correctly.

Problem 10:

(from a 2021 quiz.)

Let:

- $V \subset \mathbb{R}^5$ denote the set of solutions to the system of linear equations

$$\begin{cases} x_1 - 2x_2 + 2x_3 - x_4 + x_5 &= 0 \\ x_1 + x_2 - x_3 - x_4 - x_5 &= 2 \end{cases}$$

- And let $W \subset \mathbb{R}^5$ denote the set of solutions to the linear equation

$$2x_1 + x_2 - x_4 - 2x_5 = 0.$$

- (i) Which of V , W , $V \cap W$, $V \cup W$ are subspaces of $\mathbb{R}^5$? Briefly justify.
- (ii) If any of these are indeed a subspace, find a spanning set for it.
- (iii) Find a finite list of vectors S such that

$$V \cap W \subset \text{span}(S) \neq \mathbb{R}^5.$$

Solution to (i).

V is not a subspace because it doesn't contain the zero vector $(0, 0, 0, 0, 0)$. W is indeed a subspace because it is the set of solutions of a homogeneous system of linear equations.

$V \cap W$ is not a subspace because it does not contain the zero vector. This follows from the fact that V doesn't contain the zero vector.

$V \cup W$ isn't a subspace because it isn't closed under vector addition. After a little bit of trial and error and we can find two vectors $(4/3, 2/3, 0, 0, 0)$ and $(0, 0, 1, 0, 0)$ lying in $V \cup W$ such that their sum

$$(4/3, 2/3, 1, 0, 0)$$

doesn't lie in $V \cup W$.

To get full marks you needed to give some correct detail in each of these cases.

Solution to (ii).

The only subspace on that list is W , so we just have to find a spanning set for that.

We can immediately solve this equation by back-substitution. We introduce free parameters:

$$x_2 = p, x_3 = q, x_4 = r, x_5 = s.$$

And then solve for x_1 :

$$x_1 = \frac{1}{2}(-x_2 + x_4 + 2x_5) = -\frac{1}{2}p + \frac{1}{2}r + s.$$

A parametrization for the set of solutions is given by:

$$\left\{ \left(-\frac{1}{2}p + \frac{1}{2}r + s, p, q, r, s \right), p, q, r, s \in \mathbb{R} \right\}.$$

Separating variables we can write this equation as:

$$p(-1/2, 1, 0, 0, 0) + q(0, 0, 1, 0, 0) + r(1/2, 0, 0, 1, 0) + s(1, 0, 0, 0, 1).$$

This gives us the spanning set:

$$\{(-1/2, 1, 0, 0, 0), (0, 0, 1, 0, 0), (1/2, 0, 0, 1, 0), (1, 0, 0, 0, 1)\}.$$

A comment about student's use of notation here: A “spanning set” S for a subspace W means some set S whose span equals W . The logical equation to write is

$$\text{span}(S) = W.$$

Many students wrote $\text{span}(W)$ which is an illogical usage. Strictly speaking this means the set of vectors you can build from the vectors in W . Because W is a subspace anything you can build from W is back in W . So a correct equation (but useless in this problem) is:

$$\text{span}(W) = W.$$

There is no notation for a spanning set, because it isn't a function on W . There are many choices of spanning set. That's why we must write “a” spanning set.

Solution to (iii).

Note that $V \cap W$ isn't a subspace so a direct application of the usual procedure doesn't work, strictly speaking. To answer this question we need to think up a subspace X such that

$$V \cap W \subset X \subset \mathbb{R}^5.$$

There are actually many ways to solve this problem. The easiest solution is just to set $X = W$! And then just use your answer to Part (ii). This is a correct and immediate solution.

Another method some students used was to pick the one homogeneous equation defining V , and the one homogeneous equation defining W . In other words, to set X to be the set of solutions to the system:

$$\begin{cases} x_1 - 2x_2 + 2x_3 - x_4 + x_5 &= 0 \\ 2x_1 + x_2 - x_4 - 2x_5 &= 0. \end{cases}$$

This is a subspace containing $V \cap W$.

The most obvious solution, which was the most common correct solution was to first find a parametrization for $V \cap W$, and then to observe a spanning set from that.

To do that: The set $V \cap W$ is the set of solutions to the system of equations we get by taking both the equations in V and the equations in W :

$$\begin{cases} x_1 - 2x_2 + 2x_3 - x_4 + x_5 &= 0 \\ x_1 + x_2 - x_3 - x_4 - x_5 &= 2 \\ 2x_1 + x_2 - x_4 - 2x_5 &= 0. \end{cases}$$

To find the parametrization we solve this using Gauss-Jordan elimination:

$$\left[\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 1 & 1 & -1 & -1 & -1 & 2 \\ 2 & 1 & 0 & -1 & -2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 3 & -3 & 0 & -2 & 2 \\ 0 & 5 & -4 & 1 & -4 & 0 \end{array} \right]$$

$$\begin{aligned}
&\rightarrow \left[\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & -2/3 & 2/3 \\ 0 & 5 & -4 & 1 & -4 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccccc|c} 1 & -2 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 0 & -2/3 & 2/3 \\ 0 & 0 & 1 & 1 & -2/3 & -10/3 \end{array} \right] \\
&\rightarrow \left[\begin{array}{ccccc|c} 1 & -2 & 0 & -3 & 7/3 & 20/3 \\ 0 & 1 & 0 & 1 & -4/3 & -8/3 \\ 0 & 0 & 1 & 1 & -2/3 & -10/3 \end{array} \right] \rightarrow \left[\begin{array}{ccccc|c} 1 & 0 & 0 & -1 & -1/3 & 4/3 \\ 0 & 1 & 0 & 1 & -4/3 & -8/3 \\ 0 & 0 & 1 & 1 & -2/3 & -10/3 \end{array} \right].
\end{aligned}$$

Now we can write down the parametrization. We introduce free parameters $x_4 = r$ and $x_5 = s$ and solve:

$$\begin{cases} x_1 = 4/3 + r + 1/3s \\ x_2 = -8/3 - r + 4/3s \\ x_3 = -10/3 - r + 2/3s \\ x_4 = r \\ x_5 = s \end{cases}$$

Separating variables we can rewrite this:

$$V \cap W = \{(4/3, -8/3, -10/3, 0, 0) + r(1, -1, -1, 1, 0) + s(1/3, 4/3, 2/3, 0, 1); r, s \in \mathbb{R}\}.$$

Now we can observe from this a suitable list of vectors:

$$S = \{(4/3, -8/3, -10/3, 0, 0), (1, -1, -1, 1, 0), (1/3, 4/3, 2/3, 0, 1)\}.$$

It is clear that every vector of $V \cap W$ is a linear combination of these three vectors so that

$$V \cap W \subset \text{span}(S).$$

(To fit this into the discussion above we set $X = \text{span}(S)$.)

Furthermore S only contains 3 vectors so $\text{span}(S) \neq \mathbb{R}^5$.

□